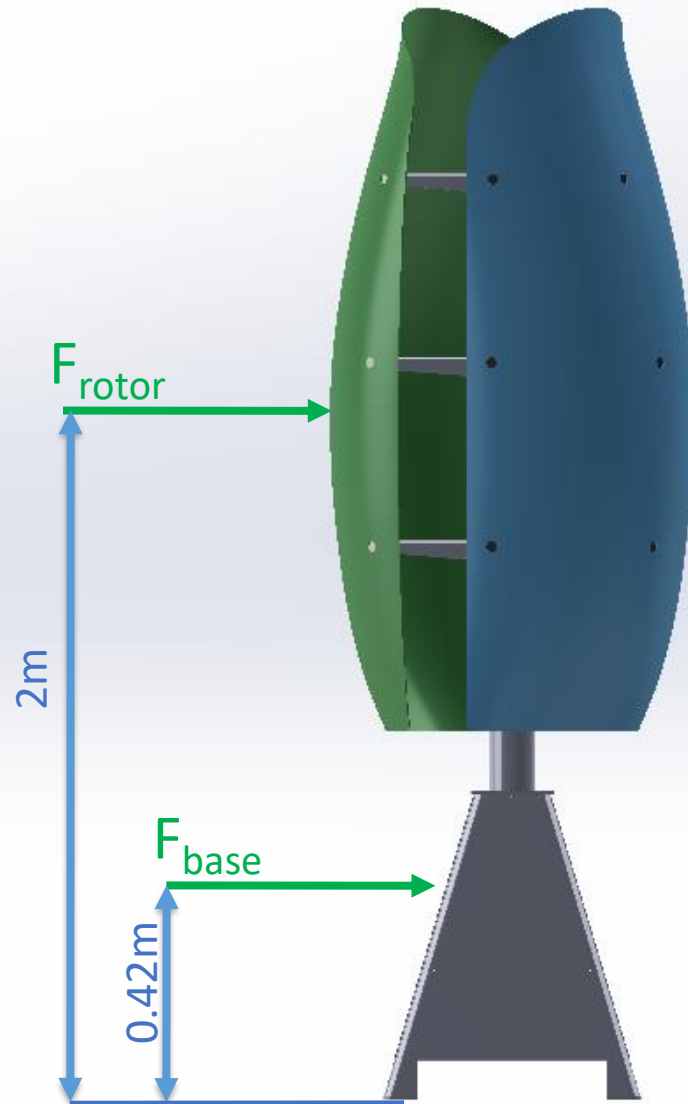


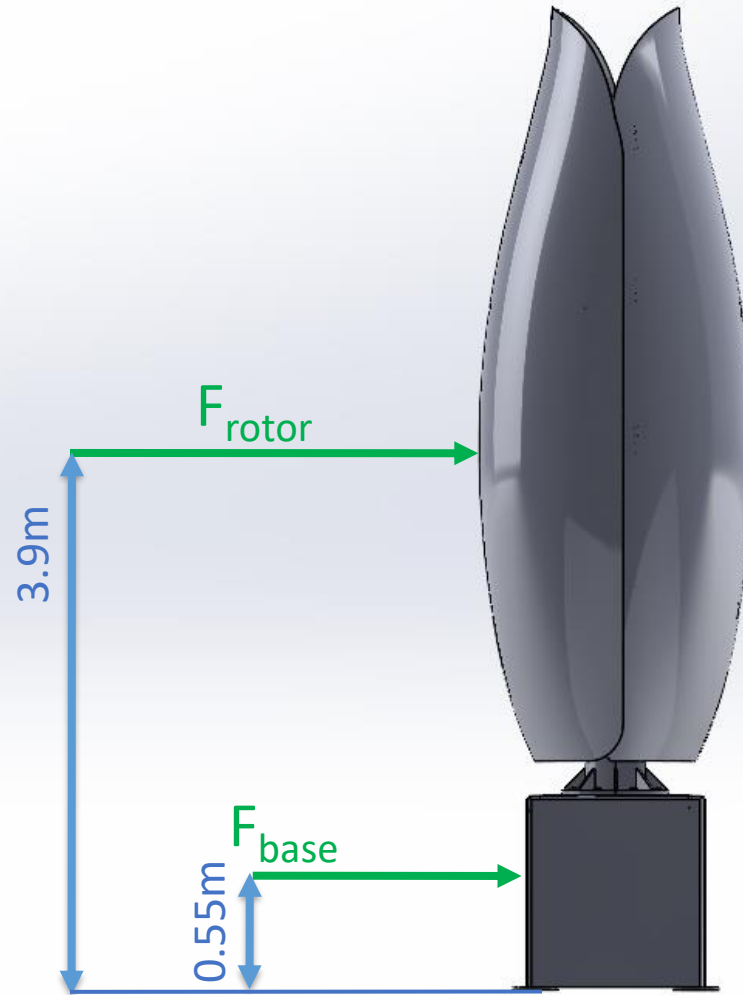
3m Flower Turbine external loads



- Reference wind speed $u_{\text{ref}} = 30 \text{ [m/s]}$
 - Wind load on rotor $F_{\text{rotor}} = 2.7 \text{ [kN]}$
 - Wind load on base $F_{\text{base}} = 0.4 \text{ [kN]}$
- Max. wind gust $u_{\text{gust}} = 42 \text{ [m/s]}$
 - Wind load on rotor $F_{\text{rotor}} = 5.3 \text{ [kN]}$
 - Wind load on base $F_{\text{base}} = 0.8 \text{ [kN]}$
- Mass $m = 300 \text{ [kg]}$

*No safety factors applied

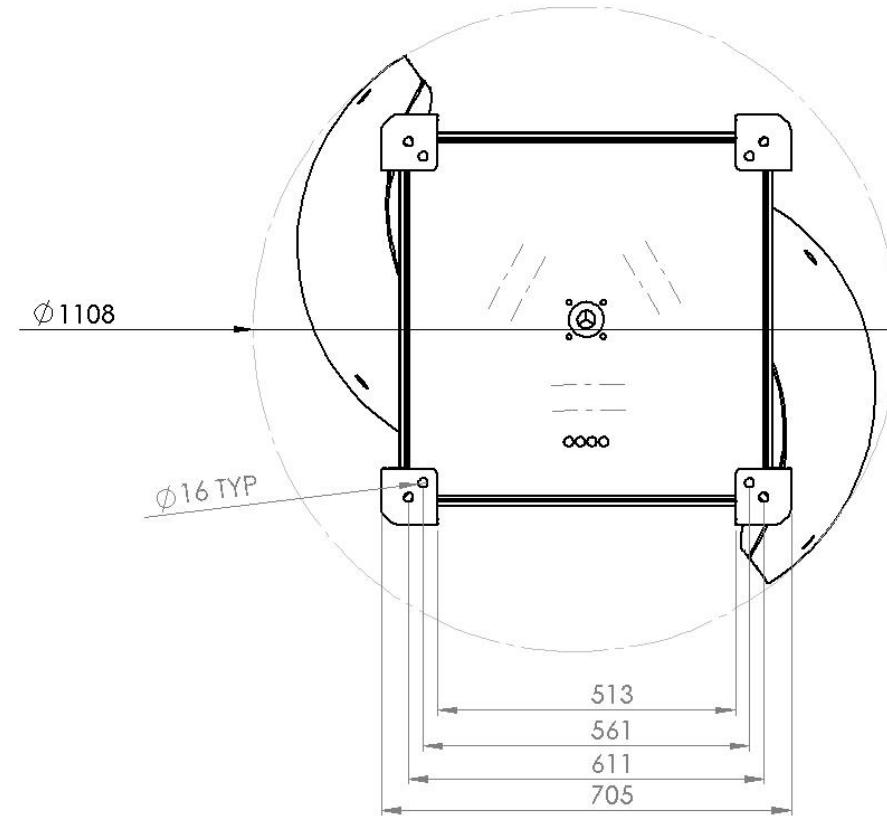
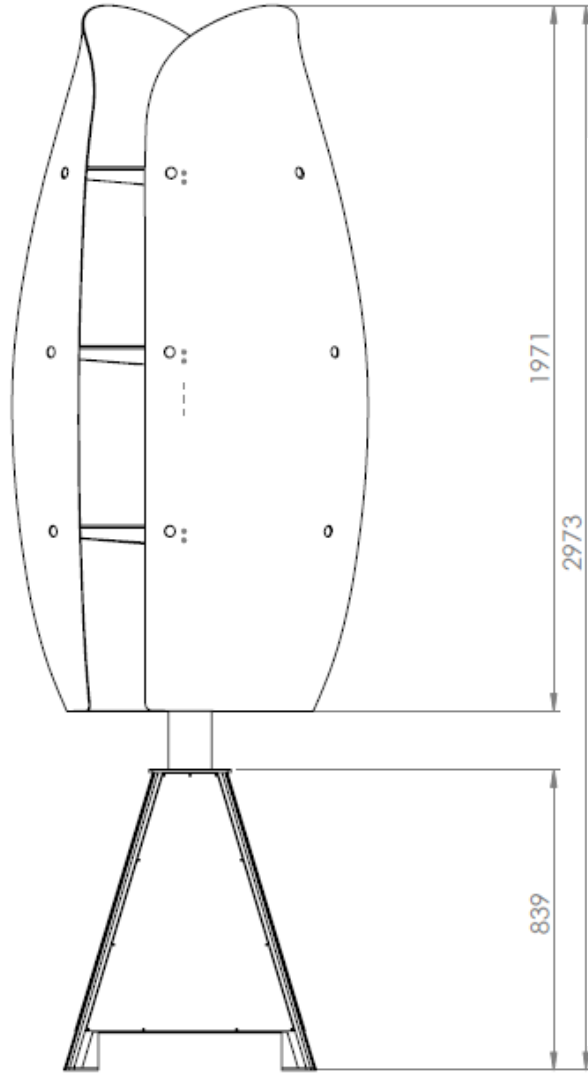
6m Flower Turbine external loads



- Reference wind speed $u_{\text{ref}} = 30 \text{ [m/s]}$
 - Wind load on rotor $F_{\text{rotor}} = 14.9 \text{ [kN]}$
 - Wind load on base $F_{\text{base}} = 0.9 \text{ [kN]}$
- Max. wind gust $u_{\text{gust}} = 42 \text{ [m/s]}$
 - Wind load on rotor $F_{\text{rotor}} = 29.1 \text{ [kN]}$
 - Wind load on base $F_{\text{base}} = 1.9 \text{ [kN]}$
- Mass $m = 4000 \text{ [kg]}$

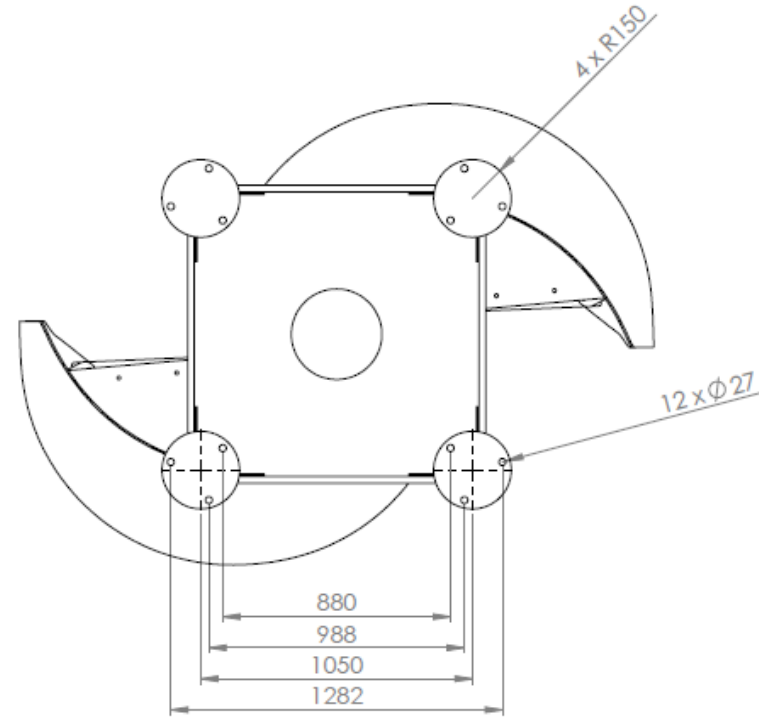
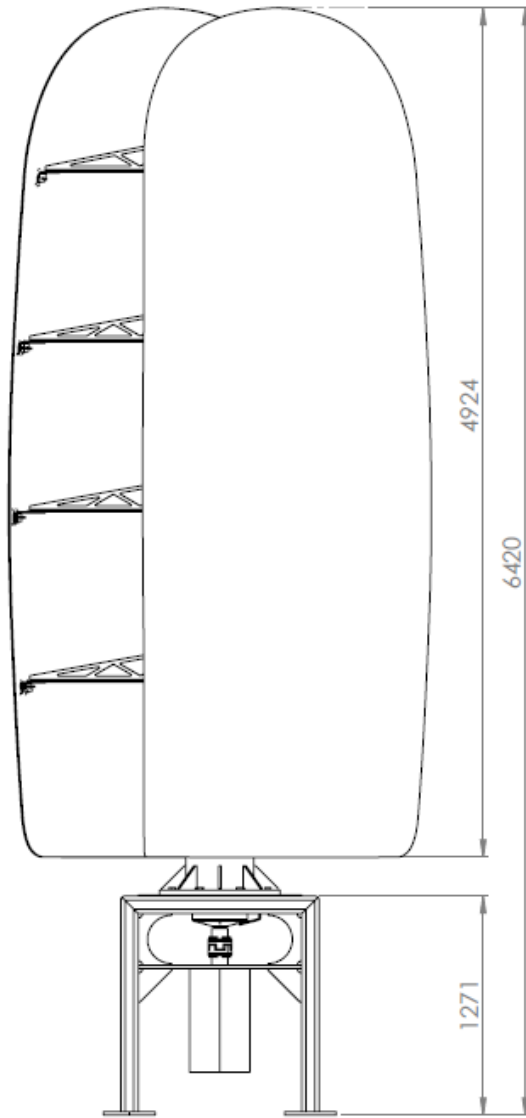
*No safety factors applied

3m Flower Turbine external dimensions



all dimenions in milimeters

6m Flower Turbine external dimensions



all dimensions in milimeters

Calculation explanation

1. First assumption is that both blades are fully loaded from the wind flow.
2. The convex blade is assumed to have a drag coefficient of 1.2
3. The concave blade is assumed to have a drag coefficient of 2.3.
4. The wind load on the rotor is represented as a concentrated force on the mid-height cross section of the rotor.
5. The base is assumed to be a rectangular box